

BCSE313L	Fundamentals of Fog and Edge Computing	L	T	P	C
		3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
1. To introduce IoT enabling technologies and its opportunities.					
2. To review underlying technologies, limitations, and challenges along with performance metrics					
and discuss generic conceptual framework in fog computing.					
3. To impart the knowledge to log the sensor data and to perform further data analytics.					
Course Outcome					
At the end of this course, student will be able to:					
1. Explore technologies behind the communication and management of fogs and edge resources.					
2. Learn the techniques for storage and computation in fogs, edges, 5G and clouds.					
3. Implement Internet of Everything (IoE) applications through fog computing architecture and use optimization techniques for the same.					
4. Analyze the performance and issues of the applications developed using fog and edge architecture.					
Module:1	Internet of Things (IoT) and New Computing Paradigms	6 Hours			
Introduction - Relevant Technologies - Fog and Edge Computing Completing the Cloud - Hierarchy of Fog and Edge Computing - Business Models – Edge Computing Platforms - Opportunities and Challenges					
Module:2	Challenges in Federating Edge Resources	6 Hours			
Introduction - Methodology - Integrated C2F2T Literature by Modeling Technique - Integrated C2F2T Literature by Use - Case Scenarios - Integrated C2F2T Literature by Metrics – Threads - Standards					
Module:3	Orchestration of Network Slices in Fog, Edge, and Clouds	6 Hours			
Introduction – Background - Network Slicing - Network Slicing in Software-Defined Clouds- Network Slicing Management in Edge and Fog - Internet of Vehicles (IoV): Architecture, Protocols and Seven-layer security model architecture for Internet of Vehicles - IoV: Network Models, Challenges and future aspects					
Module:4	Optimization Problems in Fog and Edge Computing	6 Hours			
Preliminaries - The Case for Optimization in Fog Computing-Formal Modeling Framework for Fog Computing – Metrics - Further Quality Attributes - Optimization Opportunities along the Fog Architecture - Optimization Opportunities along the Service Life Cycle - Toward a Taxonomy of Optimization Problems in Fog Computing					
Module:5	Middleware for Fog and Edge Computing	6 Hours			
Need for Fog and Edge Computing Middleware - Design Goals-State-of-the-Art Middleware Infrastructures - System Model - Case Study.					
Module:6	Technologies in Fog Computing	7 Hours			
Fog Data Management - Smart Building - Predictive Analysis with FogTorch - Machine Learning in Fog Computing - Data Analytics in the Fog - Data Analytics in the Fog Architecture.					
Module:7	Applications of Fog and Edge Computing	6 Hours			
Exploiting Fog Computing in Health Monitoring-Smart Surveillance Video Stream Processing at the Edge for Real - Time Human Objects Tracking-Fog Computing Model for Evolving Smart Transportation Applications - Testing Perspectives of Fog - Based IoT Applications - Legal Aspects of Operating IoT Applications in the Fog					